EXPERIMENT 10

Aim: Explore the Cryptocurrency Landscape

Theory:

This experiment explores the **cryptocurrency landscape** by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from **three reputable sources** as per the lab requirement:

1. **CoinGecko API** – Provided historical price data, market capitalization, and trading volumes in a structured JSON format, enabling direct processing in Python.
2. **CoinMarketCap** – Offered real-time and historical price charts, circulating supply, and market rank for visual cross-verification.
3. **Crypto.com** – Supplied quick-access market statistics, simplified price trends, and cryptocurrency news updates.

Cryptocurrency Landscape

1. **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
2. **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
    - **Store of Value** – Bitcoin, Litecoin.
    - **Utility Tokens** – Chainlink, Ethereum.
    - **Stablecoins** – USDT, USDC.
    - **Governance Tokens** – UNI, AAVE.
3. **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and regulatory updates.
4. **Data Analysis Importance** – Tracking historical trends, market capitalization, and trading volume is essential for understanding asset performance.

5. **Advancements** – Growing use of **DeFi platforms**, **NFT ecosystems**, **layer-2 scaling**, and **cross-chain protocols**.

Data Collection Process

The program used the **CoinGecko API** to automate historical data fetching for Chainlink. The retrieved data was processed with **Pandas** to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.
- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

Visualization & Dashboard

Using **Matplotlib**, a six-panel dashboard was created to display:

1. **Price Trend** – Daily price movement over the selected period.
2. **Price vs Volume** – Correlation between trading activity and price changes.
3. **Daily Returns Distribution** – Statistical spread of daily gains/losses.
4. **Moving Averages** – Technical analysis trend lines for price movement.
5. **Cumulative Returns** – Overall growth performance.
6. **Rolling Volatility** – Market risk trends over time.

Technological Analysis

Beyond market data, the study covered **advancements in blockchain technology**, including:

- **Consensus Mechanisms** – PoW (Proof of Work), PoS (Proof of Stake), and hybrid systems.
- **Scalability Solutions** – Layer-2 protocols, sharding, and sidechains to increase throughput.
- **Interoperability Protocols** – Chainlink's **CCIP**, Polkadot, and Cosmos enabling cross-chain data exchange.
- **Privacy Features** – Zero-knowledge proofs (zk-SNARKs) and ring signatures to ensure transaction confidentiality.

Chainlink's **roadmap** highlights staking implementation, advanced oracle features, and enhanced cross-chain capabilities, aligning with industry trends toward **decentralization, interoperability, and security**.

Market Trends & Adoption

- **Growing Merchant Acceptance** – More businesses accepting cryptocurrency payments.
- **Wallet Growth** – Increasing downloads and installations of crypto wallets.
- **Network Activity** – Higher transaction volumes and active addresses, signaling strong adoption.
- **Integration with Traditional Finance** – Partnerships between crypto platforms and fintech/payment providers.
- **Regulatory Developments** – Countries exploring frameworks for safe cryptocurrency adoption.

Utilizing Api for analysis and visualization :

Code:

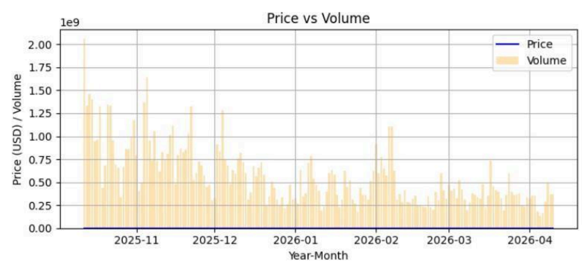
```
import requests

import matplotlib.pyplot as plt

import pandas as pd


# ------- Fetch Data -------

def get_historical_data(coin_id, days='180', currency='usd'):

    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'

    params = {'vs_currency': currency, 'days': days}

    response = requests.get(url, params=params)

    return response.json()


# ---------- Dashboard Function ----------

def chainlink_dashboard(days='180'):

    coin_id = 'chainlink'
```

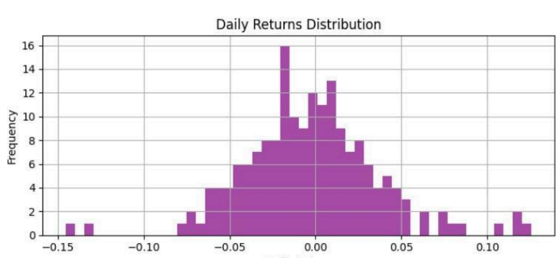
```
data = get_historical_data(coin_id, days)

# Create DataFrame
prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])
volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])

prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')
volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')

df = pd.merge(prices_df, volumes_df, on='timestamp')
df.set_index('timestamp', inplace=True)

# Calculate extra metrics
df['returns'] = df['price'].pct_change()
df['MA7'] = df['price'].rolling(window=7).mean()
df['MA30'] = df['price'].rolling(window=30).mean()
df['cumulative'] = (1 + df['returns']).cumprod()
df['volatility'] = df['returns'].rolling(window=7).std() * 100

# ---------- Plot Dashboard ----------
fig, axes = plt.subplots(3, 2, figsize=(16, 12))
fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')

# 1. Price Trend
axes[0, 0].plot(df.index, df['price'], color='blue')
axes[0, 0].set_title('Price Trend')
axes[0, 0].set_ylabel('Price (USD)')
axes[0, 0].set_xlabel('Year-Month') # Add xlabel
axes[0, 0].grid(True)
```

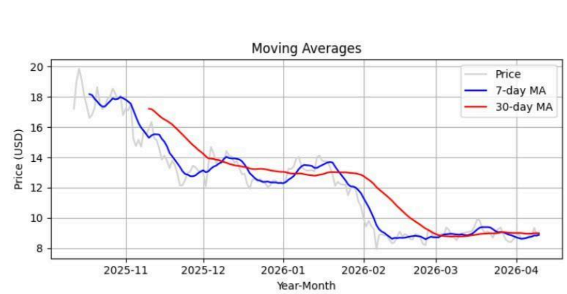
```
# 2. Price vs Volume
axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')
axes[0, 1].set_title('Price vs Volume')
axes[0, 1].set_xlabel('Year-Month') # Add xlabel
axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
axes[0, 1].legend()
axes[0, 1].grid(True)

# 3. Daily Returns Distribution
axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
axes[1, 0].set_title('Daily Returns Distribution')
axes[1, 0].set_xlabel('Daily Return')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].grid(True)

# 4. Moving Averages
axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
axes[1, 1].set_title('Moving Averages')
axes[1,  1].set_xlabel('Year-Month')
axes[1, 1].set_ylabel('Price (USD)')
axes[1, 1].legend()
axes[1, 1].grid(True)

# 5. Cumulative Returns
axes[2, 0].plot(df.index, df['cumulative'], color='green')
axes[2, 0].set_title('Cumulative Returns')
axes[2, 0].set_ylabel('Growth Factor')
```

```
    axes[2, 0].set_xlabel('Year-Month')
    axes[2, 0].grid(True)

    # 6. Volatility
    axes[2, 1].plot(df.index, df['volatility'], color='orange')
    axes[2, 1].set_title('Rolling Volatility (7-Day)')
    axes[2, 1].set_ylabel('Volatility (%)')
    axes[2, 1].set_xlabel('Year-Month')
    axes[2, 1].grid(True)

    plt.tight_layout(rect=[0, 0, 1, 0.96])
    plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing
    plt.show()
# ---------- Run Dashboard ----------
chainlink_dashboard('180')
```

Output:







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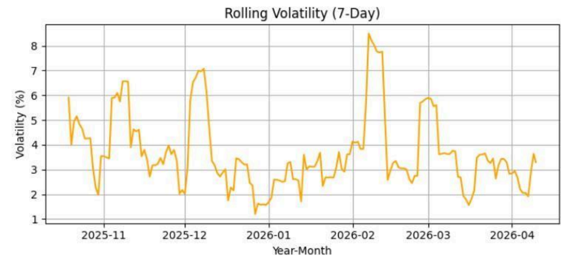
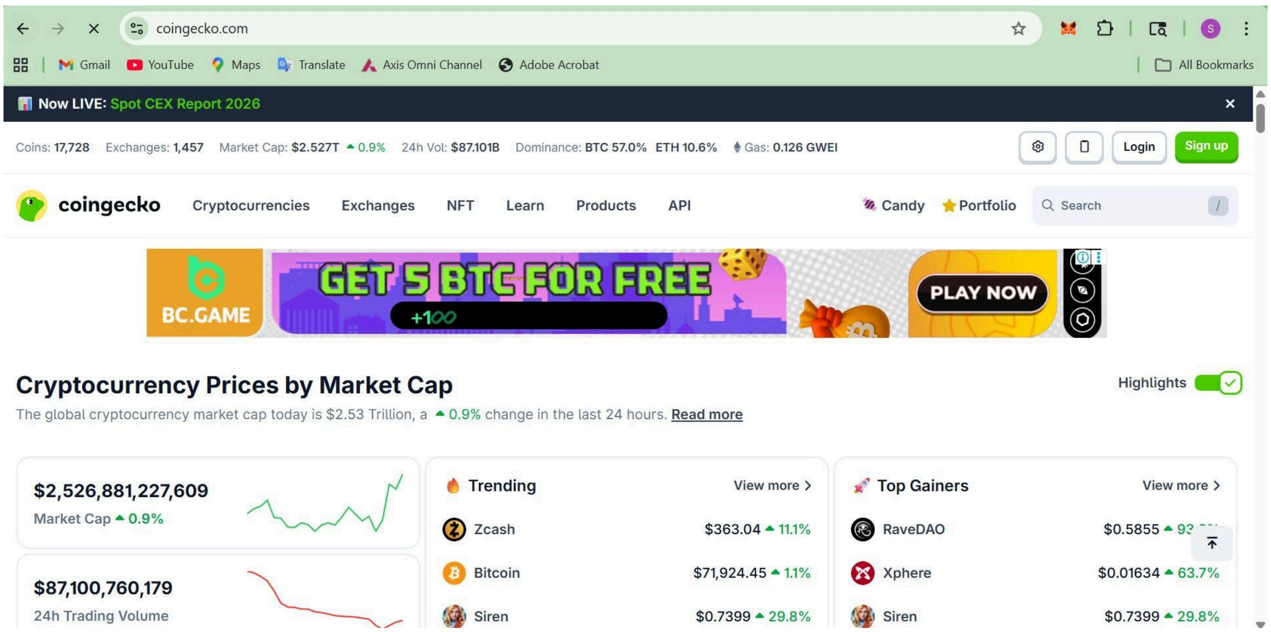
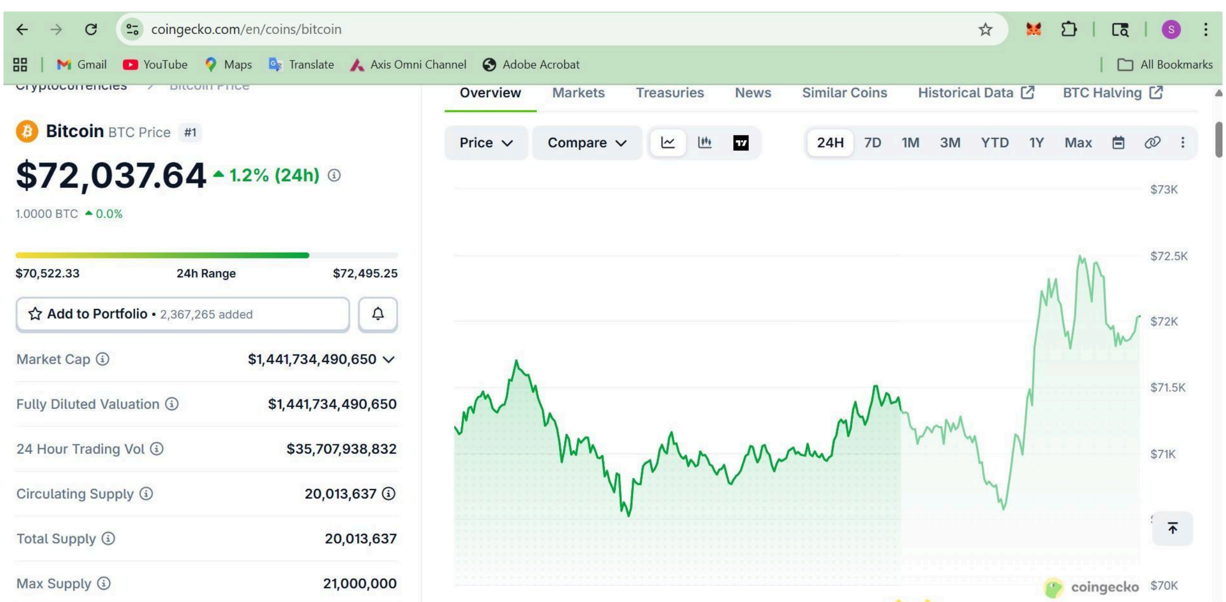


Coingecko Website





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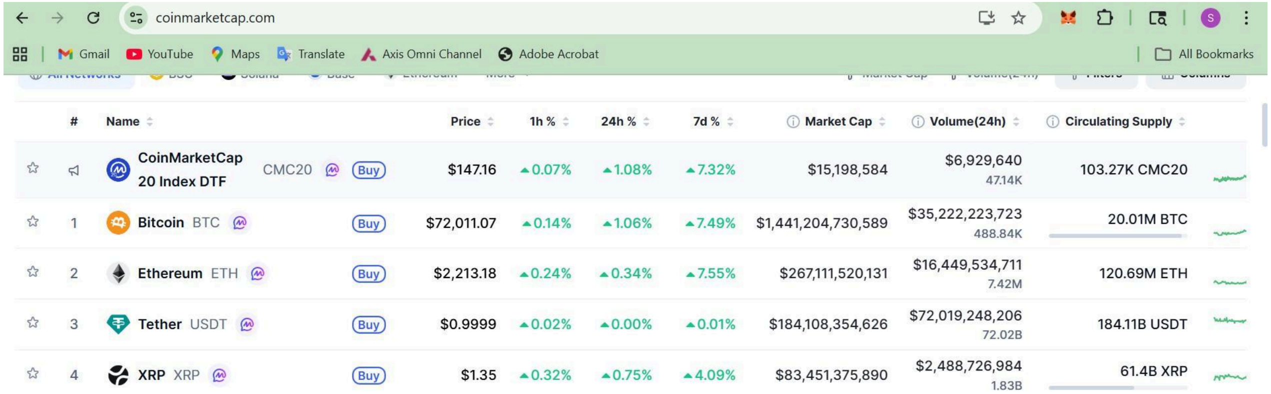
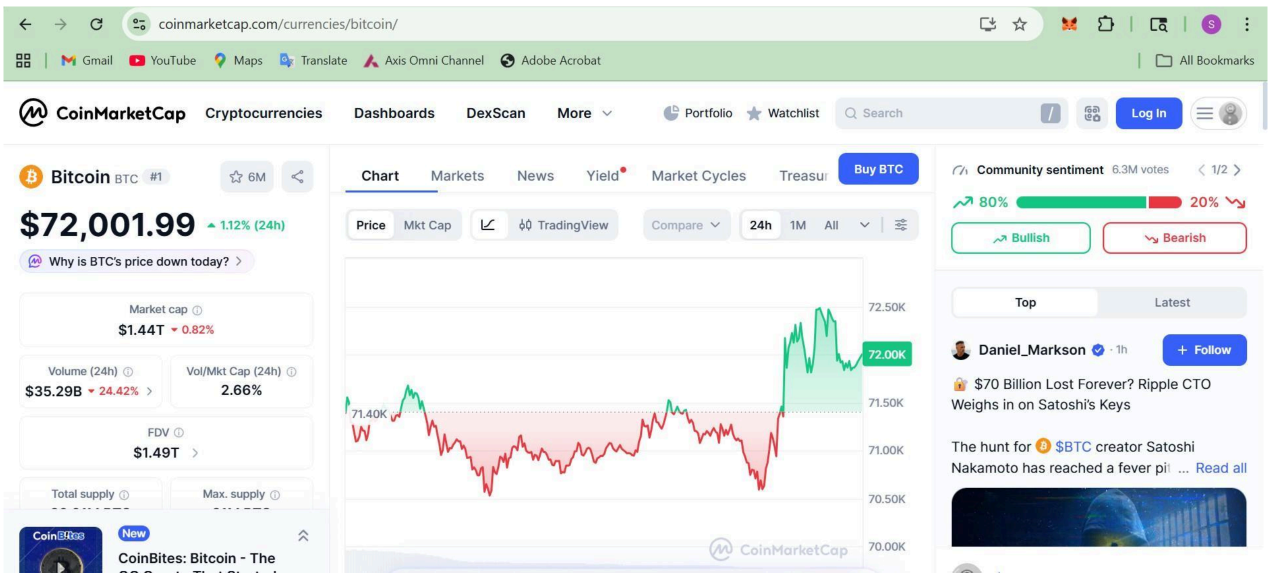
CoinMarketCap website





Conclusion:

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.